

ArduSub Controller Tuning — Surge Velocity & Yaw Rate

This document covers the ArduPilot parameters that directly affect velocity-X (surge) and yaw-rate tracking when commanding the sub in GUIDED mode via the mavlink bridge.

Pre-condition: before tuning, verify `AHRS_ORIENTATION = 6` (ROTATION_YAW_270) matches the physical Pixhawk mounting (arrow 90° CCW from vehicle forward). A 180° AHRS error will make surge and sway appear inverted and PID tuning meaningless.

1 — Surge velocity (X axis)

The bridge sends `vx` in `MAV_FRAME_BODY_FRD`. ArduSub passes it through the **Position and Speed Controller (PSC)** velocity loop before computing motor outputs.

Parameters

Parameter	Current	Recommended	Effect
<code>PSC_VELXY_P</code>	6.0	5.0	Proportional gain on velocity error. Lower → less aggressive correction, more lag. Higher → faster response, risk of oscillation.
<code>PSC_VELXY_I</code>	0.5	0.5	Integral gain. Removes steady-state error at cruise. Winds up during long runs → the source of post-stop drift.
<code>PSC_VELXY_D</code>	0.0	0.8	Derivative gain. Most impactful single change. Damps integrator wind-up oscillation. Currently 0 on real vehicle; SITL has 0.8.
<code>PSC_VELXY_FF</code>	0.2	0.0	Feedforward. Adds a bias proportional to the target. At 0.2 it introduces unnecessary thrust before any error exists.
<code>PSC_VELXY_IMAX</code>	1000	50	Integrator saturation limit (cm/s ²). At 1000 it is effectively unbounded — integrator can accumulate for seconds. Cap to 50 to bound wind-up and wind-down equally.
<code>PSC_POSXY_P</code>	1.0	2.5	Outer position loop gain. Only active when the watchdog sends a position-hold setpoint (x=y=z=0 in <code>BODY_OFFSET_NED</code>). Increase to tighten hold.
<code>PSC_VELXY_FILT_HZ</code>	—	5.0	Low-pass on velocity error fed to P/D terms. Lower → smoother but more lag. Raise only if sensor noise is low.

What each term does in practice

```

accel_cmd = P*(v_target - v_actual)
            + I*∫(v_target - v_actual) dt      ← wind-up source
            + D*d(v_target - v_actual)/dt      ← wind-down damper
            + FF*v_target                      ← bias, bypass error loop

```

- **Lag after stop** → `PSC_VELXY_IMAX` too high or `PSC_VELXY_D = 0`
- **Overshoot / backward drift** → same root cause, I winds up, D can't damp it
- **Oscillation during cruise** → `PSC_VELXY_P` too high
- **Slow ramp-up** → `PSC_VELXY_P` too low or `PSC_VELXY_FF = 0`

Tuning procedure

1. Apply `PSC_VELXY_IMAX = 50` and `PSC_VELXY_D = 0.8` first — these are parameter-only, no reflash, and have the largest effect on the lag symptom.
2. Run `./scripts/cmd_vel_ramp.py ramp surge 0.20 --ramp 5 --hold 5` and record a bag (`/pixhawk/cmd_vel + /odometry/filtered/local`).
3. Plot `twist.linear.x` vs the commanded value. Target: tracks within ± 0.05 m/s at steady state, no backward overshoot after stop.
4. If oscillation appears at steady state → reduce `PSC_VELXY_P` by 0.5 steps.
5. If tracking is sluggish → increase `PSC_VELXY_P` or add `PSC_VELXY_FF = 0.1`.
6. If post-stop drift persists → further reduce `PSC_VELXY_IMAX` (try 30).

2 — Yaw rate

The bridge sends `yaw_rate` (rad/s). ArduSub passes it through the **Attitude Controller (ATC)** rate loop.

Note: yaw direction is currently inverted in the firmware mixer (MOT_2 and MOT_6 yaw factors have wrong sign). Until the mixer is reflashed, the bridge applies `yaw_rate = -msg.angular.z` to compensate. Tuning the rate loop is still valid — the sign inversion is downstream of the PID.

Parameters

Parameter	Current	Recommended	Effect
<code>ATC_RAT_YAW_P</code>	0.08	0.18	Rate loop proportional. At 0.08 the loop barely responds — causes the chatter observed in bags.
<code>ATC_RAT_YAW_I</code>	0.0	0.018	Integrates away steady-state yaw error (drag asymmetry, thruster bias).
<code>ATC_RAT_YAW_D</code>	0.0	0.0	Keep 0 for yaw — yaw rate sensors are noisy; D amplifies this. Only add if overshoot is severe after raising P/I.
<code>ATC_RAT_YAW_FF</code>	0.0	0.10	Feedforward on yaw rate target. Eliminates the "start delay" where the sub pauses before

Parameter	Current	Recommended	Effect
			beginning to yaw.
ATC_RAT_YAW_IMAX	0.222	0.222	Already correct — leave as-is.
ATC_RAT_YAW_FLTT	—	10.0	Low-pass on the rate target. Smooths step commands.
ATC_RAT_YAW_FLTE	—	10.0	Low-pass on the rate error fed to P/I.
ATC_ANG_YAW_P	0.0	0.0	Heading-hold outer loop. Intentionally 0 — pure rate control, no heading reference needed in GUIDED velocity mode.
ATC_SLEW_YAW	6000	200	Max yaw acceleration (deg/s ²). At 6000 this is a step — causes the chatter spike at onset. 200 is a realistic ramp.
ATC_RATE_Y_MAX	180	90	Hard cap on commanded yaw rate (deg/s). 180 is too fast for this hull.

What each term does in practice

```
torque_cmd = P*(ω_target - ω_actual)
            + I*∫(ω_target - ω_actual) dt
            + D*d(ω_target - ω_actual)/dt    ← keep 0, sensor noise
            + FF*ω_target
```

- **Chatter / ringing at onset** → **ATC_SLEW_YAW** too high (step input) or **ATC_RAT_YAW_P** too high
- **Sub barely yaws** → **ATC_RAT_YAW_P** too low (0.08 is the current symptom)
- **Yaw never reaches commanded rate** → **ATC_RAT_YAW_I** = 0 (no integrator to close SS error)
- **Start delay before yaw begins** → **ATC_RAT_YAW_FF** = 0

Tuning procedure

1. Apply **ATC_SLEW_YAW** = 200 and **ATC_RATE_Y_MAX** = 90 first — reduces the onset spike regardless of PID values.
2. Set **ATC_RAT_YAW_P** = 0.18, **ATC_RAT_YAW_I** = 0.018, **ATC_RAT_YAW_FF** = 0.10.
3. Run `./scripts/cmd_vel_ramp.py ramp yaw 0.30 --ramp 4 --hold 4` and record `/imu/angular_velocity + /pixhawk/cmd_vel`.
4. Plot `angular_velocity.z` vs commanded rate. Target: tracks within ±0.05 rad/s, no ringing, no >0.1 rad/s overshoot.
5. If ringing persists → reduce **ATC_RAT_YAW_P** in 0.02 steps.
6. If response is sluggish → increase **ATC_RAT_YAW_FF** in 0.05 steps first (FF is cleaner than P for this).

3 — Recommended param block (apply all at once)

Paste into MAVProxy console or load as **.parm** file. Reboot FC after.

```
param set PSC_VELXY_P      5.0
param set PSC_VELXY_D      0.8
param set PSC_VELXY_FF     0.0
param set PSC_VELXY_IMAX   50
param set PSC_POSXY_P      2.5

param set ATC_RAT_YAW_P     0.18
param set ATC_RAT_YAW_I     0.018
param set ATC_RAT_YAW_FF    0.10
param set ATC_SLEW_YAW     200
param set ATC_RATE_Y_MAX    90
param set MOT_SAFE_DISARM   1
```

4 — Test profiles

All profiles publish to `/pixhawk/cmd_vel` at 20 Hz.

```
# Surge – trapezoid, good baseline for lag measurement
./scripts/cmd_vel_ramp.py ramp surge 0.20 --ramp 5 --hold 5

# Surge – bipolar, catches asymmetric thrust (MOT_1 direction check)
./scripts/cmd_vel_ramp.py bipolar surge 0.20 --ramp 3 --hold 2

# Surge – staircase, characterises tracking vs amplitude (finds deadband)
./scripts/cmd_vel_ramp.py staircase surge 0.30 --hold 3

# Yaw rate – trapezoid, primary ATC_RAT_YAW tuning run
./scripts/cmd_vel_ramp.py ramp yaw 0.30 --ramp 4 --hold 4

# Yaw rate – bipolar, exposes direction asymmetry
./scripts/cmd_vel_ramp.py bipolar yaw 0.30 --ramp 3 --hold 2
```

Record with:

```
ros2 bag record -s mcap -o ~/Downloads/Rosbags/$(date +%Y%m%d_%H%M%S)_tune \
  /pixhawk/cmd_vel /odometry/filtered/local /imu/angular_velocity \
  /filter/euler /pixhawk/servo_output_raw
```